

Kushal Kumar Rajak

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Summary

Backend Engineer with 4+ years of experience building scalable microservices, distributed systems, and cloud-native applications using Node.js, NestJS, TypeScript, and AWS. Experienced in designing event-driven architectures, high-performance APIs, Kafka-based messaging systems, Redis caching layers, and enterprise-grade backend platforms.

Skills

Backend: Node.js, NestJS, TypeScript, JavaScript, Express.js, Java
Frontend: React (Basic), HTML5, CSS3
Databases: MongoDB, PostgreSQL, MySQL, Redis
Messaging: Apache Kafka, RabbitMQ
Cloud & DevOps: AWS, Docker, Kubernetes, Jenkins, GitLab CI/CD
Architecture: Microservices, Distributed Systems, Event-Driven Architecture, CQRS
AI Technologies: LangChain, RAG, LLM Integration, Agentic AI
Tools: Git, Linux, Shell Scripting

Experience

Software Development Engineer — Ginesys Mar 2025 – Present

- Developed scalable microservices and REST APIs using NestJS, TypeScript, PostgreSQL, and Redis.
- Designed event-driven architectures using Kafka for asynchronous communication between services.
- Built distributed inventory and order-processing systems handling millions of records.
- Implemented Redis-based caching, rate limiting, and session management to improve application performance.
- Developed CI/CD pipelines using Jenkins and GitLab CI for automated deployments.
- Containerized applications using Docker and deployed cloud-native workloads on AWS.
- Optimized database queries and batch-processing workflows, significantly reducing execution time.
- Collaborated with cross-functional teams to deliver enterprise-scale backend solutions.

Software Development Engineer — Emiza Supply Chain Services Feb 2024 – May 2025

- Developed backend services using NestJS, Node.js, TypeScript, and PostgreSQL.
- Designed and implemented secure REST APIs with authentication and authorization mechanisms.
- Integrated Kafka-based asynchronous messaging for inventory and warehouse operations.
- Improved application performance through Redis caching and database query optimization.
- Built containerized deployments using Docker and automated CI/CD pipelines.
- Participated in designing scalable microservice architectures for supply chain applications.

Software Engineer — Diatoz Solutions Sep 2021 – Feb 2024

- Developed scalable backend applications using Node.js, NestJS, TypeScript, and MongoDB.
- Built REST APIs and integrated third-party services for enterprise applications.
- Implemented authentication, authorization, and API security best practices.
- Developed reusable backend modules and shared service components.
- Improved system performance through caching and database optimization techniques.
- Automated deployment workflows using Linux shell scripting and DevOps practices.

Projects

Multi-Tenant Data Processing Platform

- Designed and developed a highly scalable NestJS microservices platform for bulk CSV processing.
- Implemented Kafka-based asynchronous event processing and workflow orchestration.
- Used Redis for distributed caching, queue management, and rate limiting.
- Built tenant-aware architecture supporting dynamic validation rules and configurations.
- Optimized batch processing to support high-volume data ingestion.

Distributed Code Execution Platform

- Designed a secure and scalable code execution platform for running untrusted code.
- Built microservices using NestJS, Kafka, Redis, and PostgreSQL.
- Implemented job scheduling, retry mechanisms, timeout handling, and fault recovery.
- Developed scalable execution workers supporting multiple programming languages.
- Designed distributed architecture capable of horizontal scaling.

Event-Driven Order Processing System

- Developed Kafka-based producer-consumer architecture for order processing workflows.
- Implemented retry mechanisms, dead-letter queues, and message durability strategies.
- Built fault-tolerant services capable of handling high transaction volumes.
- Improved throughput and reduced processing latency under peak load conditions.

Real-Time Inventory Management System

- Developed inventory synchronization services using NestJS, Kafka, PostgreSQL, and Redis.
- Implemented bulk inventory adjustment processing handling millions of inventory records.
- Optimized database performance using batching, indexing, and caching strategies.
- Reduced processing latency and improved inventory synchronization reliability.

Database Design & Storage Engine

- Designed and implemented with efficient persistence and retrieval mechanisms.
- Built B-Tree based indexing for logarithmic-time search, insert, and delete operations.
- Developed transaction management supporting ACID principles.
- Implemented write-ahead logging (WAL) for durability and crash recovery.
- Optimized storage and query execution using memory caching and indexing strategies.

Education

Master of Computer Applications (MCA)

- Jain University 2025 – Present

Bachelor of Computer Applications (BCA)

- Jamshedpur Co-operative College 2019